



Ra-02 Specifications

Version V1.0

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Document Preparation/Revision/Abolition Record

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1. Product Overview

The Essence LoRa series module (Ra-02) is designed and developed by Essence Technology. This module is used for ultra-long distance spread spectrum Communication, its RF chip SX1278 mainly uses LoRa™Long-range modem for very long-distance spread spectrum communications, Strong anti-interference ability, can minimize current consumption.™Patented modulation technology, SX1278 has a high sensitivity of over -148dBm, a power output of +20dBm, a long transmission distance and high reliability. Compared with traditional modulation technology, LoRa™Modulation technology also has obvious advantages in anti-blocking and selection, solving the Traditional design solutions cannot take into account the issues of distance, anti-interference and power consumption at the same time.

Application areas: Automatic meter reading, home building automation, security systems, remote irrigation systems.

characteristic

- LoRa™modem
- Support FSK, GFSK, MSK, GMSK, LoRa™And OOK modulation
- Support frequency band 410MHz~525MHz
- The operating voltage is 3.3V, the maximum output is +20dBm, and the maximum operating current is 105mA
- It has low power consumption in receiving state, with receiving current of 12.15mA and standby current of 1.6mA
- High sensitivity: as low as -140dBm
- Small size double row stamp hole patch package
- The module uses SPI interface, half-duplex communication, CRC, up to 256 bytes of data packet engine

Main parameters

Table 1 Description of main parameters

Module Model	Ra-02
Encapsulation	SMD-16
size	17*16*3.2(± 0.2)MM
Antenna type	IPEX socket
Spectrum range	410MHz~525MHz
Operating temperature	-40°C ~ 85°C
Storage environment	- 40 °C ~ 125 °C , < 90%RH
Power supply range	2.7~3.6V, typical value 3.3V, current greater than 200mA
Supported interfaces	SPI
Programmable bit rate	Up to 300kbps
Certification	REACH、RoHS

2. Electrical parameters

Electrical Characteristics

parameter	name	Minimum	Typical Value	Maximum	unit
Operating temperature	TOPR	- 40	25	85	°C
Supply voltage	VDD	2.7	3.3	3.6	V

Digital Port Characteristics

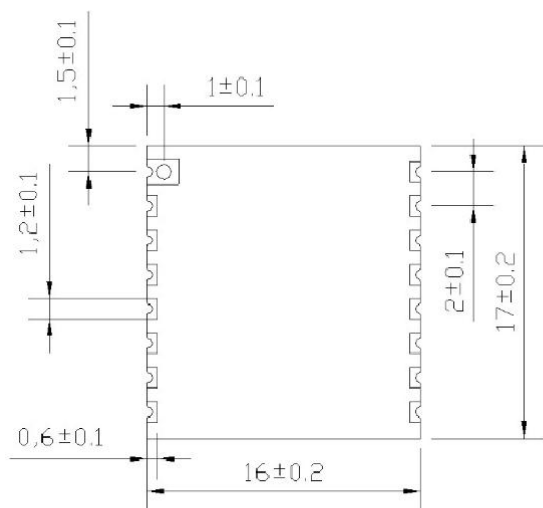
describe		Typical Value		unit	
Operating frequency		410~525		MHz	
port	name	Minimum	Typical Value	Maximum	unit
IO Level	VIO	2.7	3.3	3.6	V
Input logic level low	VIL	-	-	0.2	V
Input logic level high	VIH	0.8	-	-	V
Output logic level low	VOL	-	-	0.1	V
Output logic level high	VOH	0.9	-	-	V

SPI interface characteristics

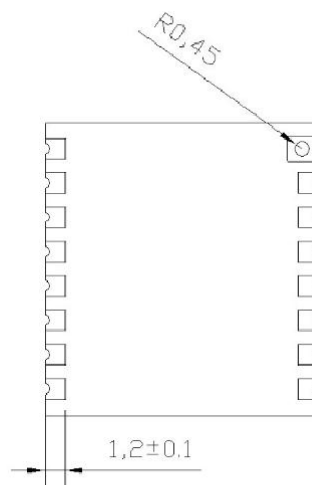
symbol	describe	condition	Minimum	typical value	most big value	unit
Fsck	SCK frequency	-	-	-	10	MHz
tch	SCK high time	-	50	-	-	ns
tcl	SCK low time	-	50	-	-	ns
trise	SCK rise time	-	-	5	-	ns
tfall	SCK Fall Time	-	-	5	-	ns

tsetup	MOSI setup time	Change from MOSI to SCK rising edge	30	-	-	ns
thold	MOSI hold time	From SCK rising edge to MOSI changes	20	-	-	ns
tnsetup	NSS establishment time	From NSS falling edge to SCK rising edge	30	-	-	ns
tnhold	NSS maintenance time	From SCK falling edge to NSS rising edge, normal model	100	-	-	ns
tnhigh	spi access interval NSS high level time	-	20	-	-	ns
T_DATA	DATA maintenance and establishment time	-	250	-	-	ns

3. Appearance and Dimensions



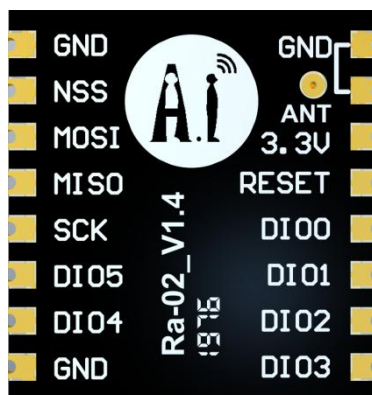
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4. Pin Definition

The Ra-02 module has 16 interfaces in total, as shown in the pin diagram. The pin function definition table is the interface definition.



Ra-02 Pin Diagram

Table pin function definition

Footer	name	Functional Description
Test Points	ANT	Connect the antenna
1	GND	Grounding
2	GND	Grounding
3	3.3V	Typical 3.3V power supply
4	RESET	Reset pin
5	DIO0	Digital IO0 software configuration
6	DIO1	Digital IO1 Software Configuration
7	DIO2	Digital IO2 Software Configuration
8	DIO3	Digital IO3 Software Configuration
9	GND	Grounding
10	DIO4	Digital IO4 Software Configuration
11	DIO5	Digital IO5 Software Configuration
12	SCK	SPI clock input
13	MISO	SPI Data Output

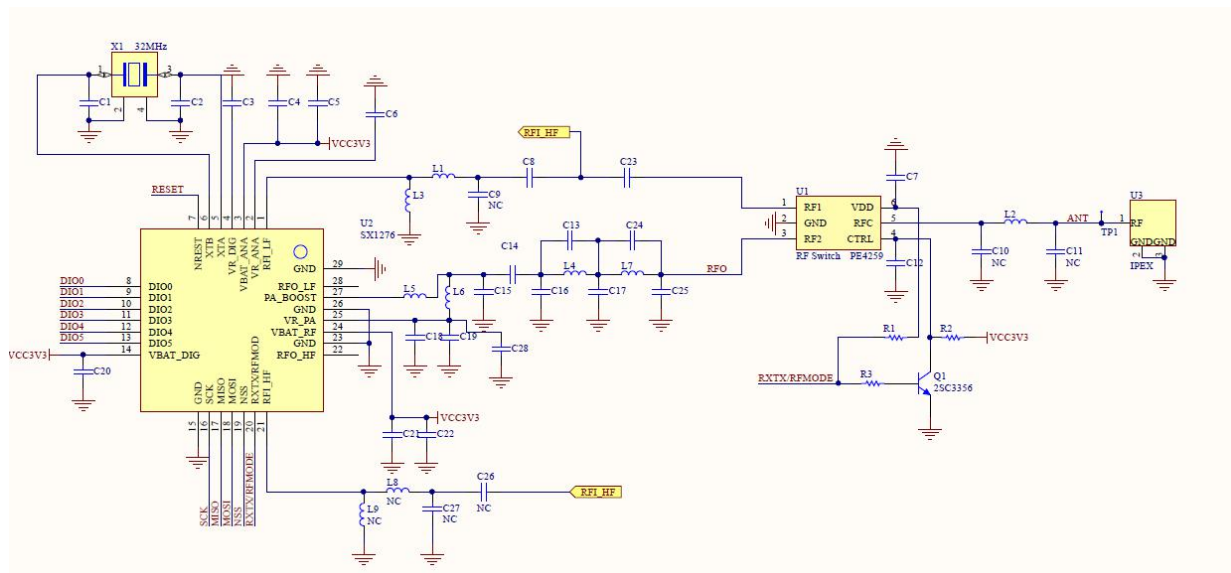
14	MOSI	SPI Data Input
15	NSS	SPI chip select input
16	GND	Grounding

SX1278's 6 general IO pins are used in LoRa™. Their mapping relationship depends on the configuration of the two registers RegDioMapping1 and RegDioMapping2.

Set.

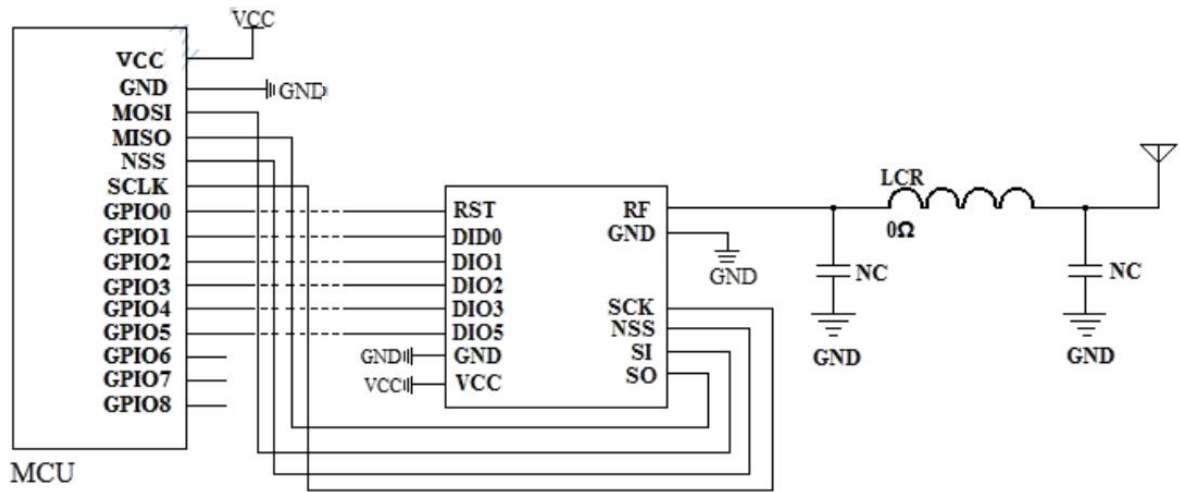
operate model	DIOx Mapping	DIO5	DIO4	DIO3	DIO2	DIO1	DIO0
all	00	ModeReady	CadDetected	CadDone	F Change Channel	RxRimeout	RxDone
	01	ClkOut	PLLlock	Valid Header	F Change Channel	F Change Channel	TxDone
	10	ClkOut	PLLlock	PayloadCrc Error	F Change Channel	CadDetected	CadDone
	11	-	-	-	-	-	-

5. Schematic diagram

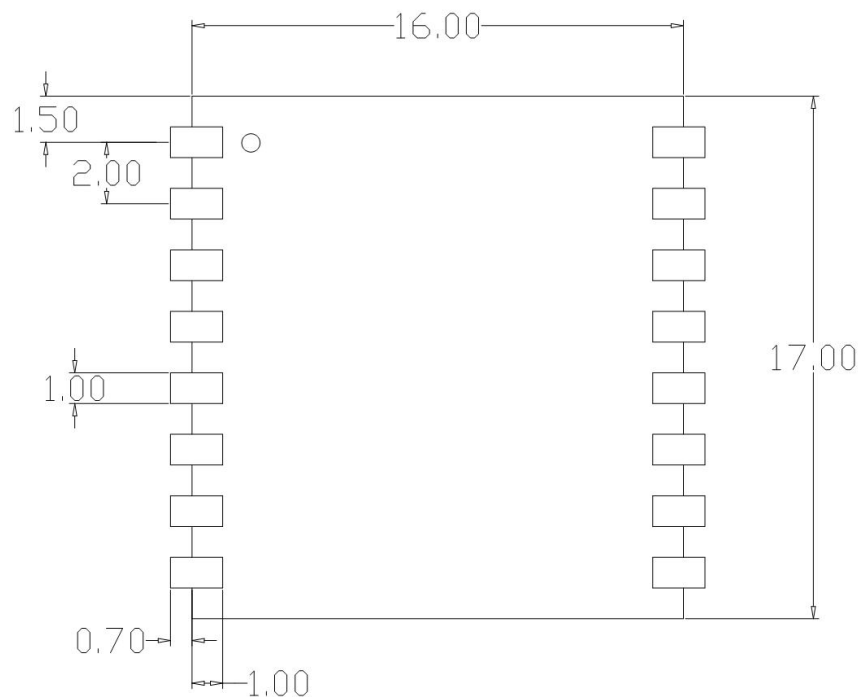


6. Design Guidance

1. Application circuit



2. Recommended module package design dimensions



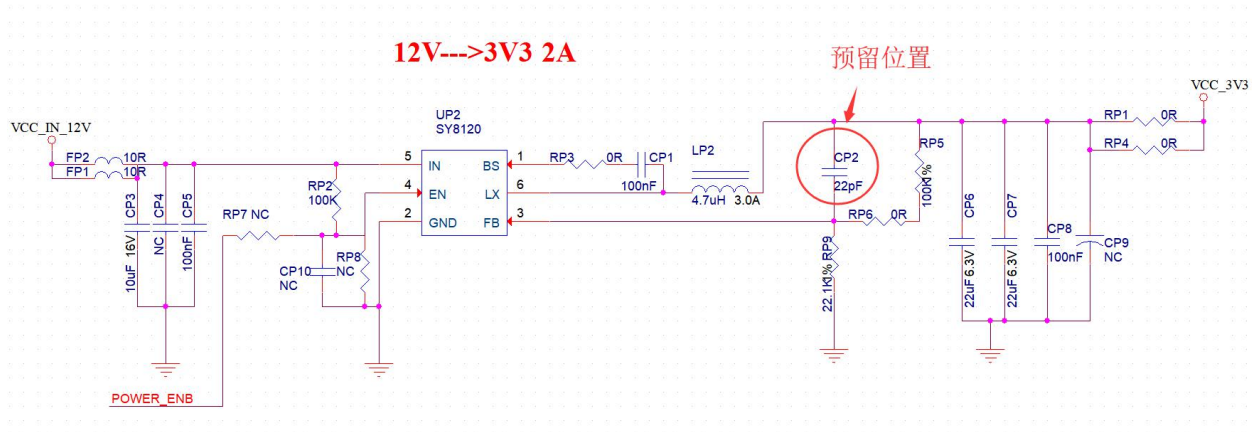
Notice: This is the Ra-02 module package diagram. It is recommended to design the PCB board according to this diagram so that the module can work normally on the PCB board. And when designing the pads, please note that the pads on the PCB cannot be designed to be offset inward compared to the corresponding pads on the module. If the PCB pads are expanded relative to the module pads, it will not affect the use of the module.

3. Antenna installation

(1) Ra-02 needs to weld the antenna. The module is compatible with both half-hole pads and round-hole pads. (2) In order for the antenna to achieve the best effect, the antenna assembly position should be away from metal parts and high-frequency devices.

4. Power supply

- (1) Recommended voltage: 3.3V, peak current: 200mA or above
- (2) It is recommended to use LDO power supply; if DC-DC is used, it is recommended to control the ripple within 30mV.
- (3) It is recommended to reserve space for dynamic response capacitors in the DC-DC power supply circuit to optimize the output ripple when the load changes greatly.
- (4) It is recommended to add ESD devices to the 3.3V power interface.



5. Use of GPIO port

- (1) Some GPIO ports are connected to the module. If you need to use them, it is recommended to connect a 10-100 ohm resistor in series to the IO port. This can suppress overshoot and make the levels on both sides more stable. It is helpful for EMI and ESD.
- (2) For the pull-up and pull-down of special IO ports, please refer to the instructions in the specification, which will affect the startup configuration of the module. (3) The IO port of the module is 3.3V. If the IO level of the main control and the module does not match, it is necessary to add a level conversion circuit. (4) If the IO port is directly connected to the peripheral interface, or terminals such as pin headers, it is recommended to reserve ESD devices near the terminals of the IO routing.

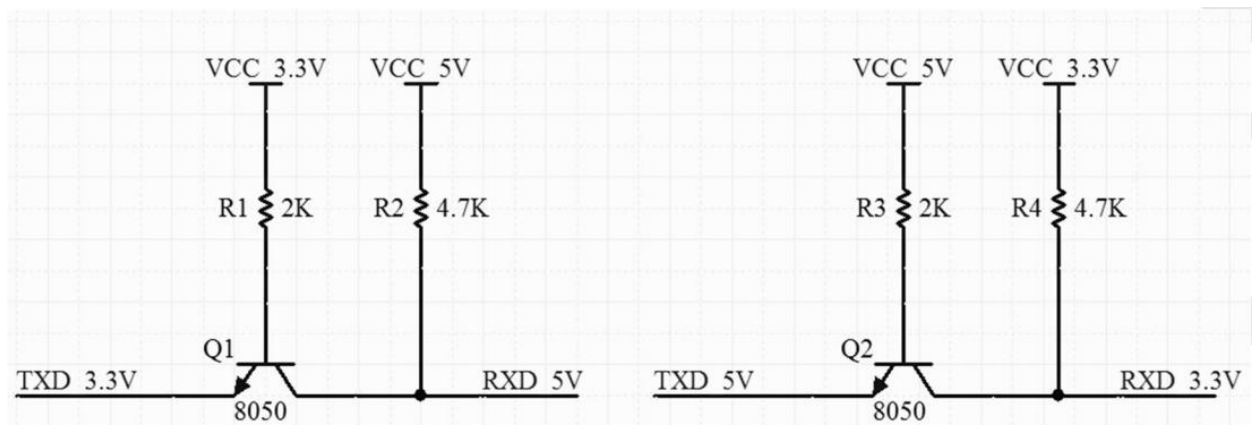
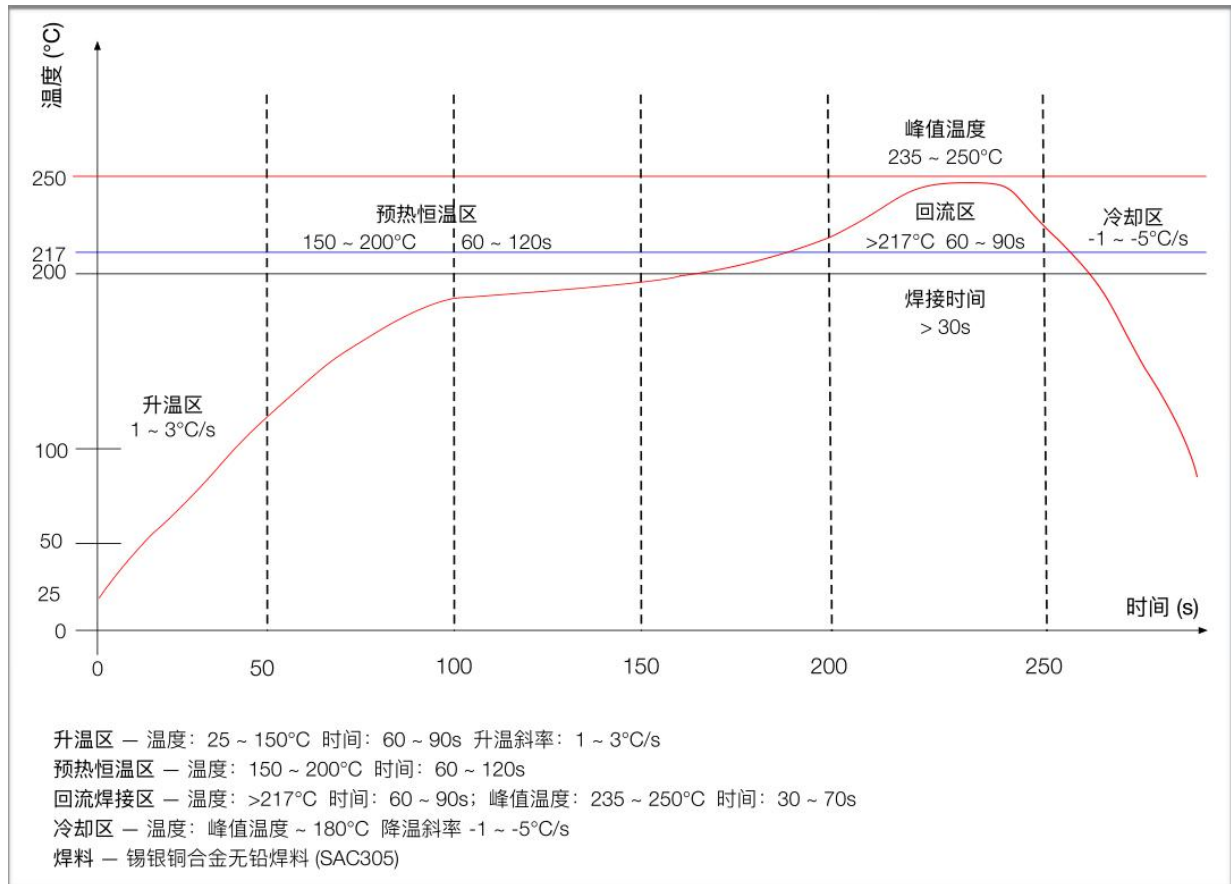


Figure 1. Level conversion circuit

7. Reflow Oven Curve



8. Packaging Information

As shown below, Ra-02 The packaging is taping.



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Official website: <https://www.ai-thinker.com>

Development Docs: <https://docs.ai-thinker.com>

Official Forum: <http://bbs.ai-thinker.com>

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